

Homework 7

due 29.11.2023, 14:00

Exercise 7.1 - Fox's Algorithm (10 points)

Write a program that reads two matrices from files, multiplies them in parallel and prints the result to a file again. Implement the Fox algorithm from the lecture, that divides square $n \times n$ matrices onto $n_{\text{proc}} = q \times q$ blocks of size $\bar{n} \times \bar{n}$, with $\bar{n} = n/q$. The first row of a matrix file contains m and n , the number of rows and columns of a matrix A . The remaining lines contain one matrix element each, in the order $a_{00}, a_{01}, \dots, a_{0\,n-1}, a_{10}, \dots, a_{m-1\,n-1}$.

- Process 0 should be responsible for all I/O. This is more challenging now than with last week's block-row-layout!
- All matrices can be assumed to be square. Every process should dynamically allocate space for its $\bar{n} \times \bar{n}$ portion of the three matrices. No global sized matrices or vectors are allowed (neither $n \times n$ nor $n \times 1$). The easiest is to use 1D arrays, such that e.g. local matrix element a_{ij} is accessed via `a[j+n*i]`.
- To test your program, use the two matrices `A.txt` and `B.txt` that you can find in `/work/korzec/LAB2/ex7/` on `stromboli` (avoid making copies, these files can be accessed from the compute nodes for reading). Your program should create the output `C.txt`, which can be visualized as an image with the matlab script `check.m`. Include this image in your PDF.
- The test matrices have $n = 1024$, so $n_{\text{proc}} = 4, 16, 64$ are reasonable choices.